

Xiangnan Feng

Postdoctoral Fellow

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Research Fields

Computational Social Science, Complex Network, Machine Learning, Data Mining, Statistical Physics

Experience

Complexity Science Hub, Vienna, Austria

2023 - Now

Postdoctoral Fellow, Transforming Economies

Supervisor: Frank Neffke (Group Leader)

Max Planck Institute for Human Development, Berlin, Germany

2021 - 2022

Postdoctoral Fellow, Future of Work

Supervisors: Iyad Rahwan (Professor), Alex Rutherford (Senior Research Scientist)

Education

Beihang University, Beijing, China

2014 - 2020

Ph.D. in Mathematics

Supervisor: Zhiming Zheng (Academician of Chinese Academy of Sciences), Wei Wei (Associate Professor)

Beihang University, Beijing, China

2010 - 2014

B.S. in Mathematics

Academic Topics

Future of Work

We study the labour market in a perspective of complexity for various topics in economics including skill evolution, automation, employment, et al.

Graph Neural Networks

We build new graph neural network models to solve network problems including link prediction, graph classification.

Network Theory

We use information theory to study fundamental problems in network like centrality, Epidemiology, et al.

Game Theory on Networks

We study game theory on networks with various dynamic strategies

Minimum Vertex Cover Problem

We study the network structure features in minimum vertex cover problem, one of the classical NP-hard problems.

Publications

Evolutionary dynamic of the coupled awareness-epidemic networks with higher-order structure

2025

Xiangyu Meng, Wei Wei, **Xiangnan Feng**, Zhenyu Shi, Baifeng Li

Physica A: Statistical Mechanics and its Applications 660, 130210

Probabilistic Time Series Forecasting via Diffusion Models under Multi-Scale Guidance

2024

Bowen Pang, Liyi Huang, **Xiangnan Feng**, Wei Wei
Available at SSRN 5071353

- Link prediction by continuous spatiotemporal representation via neural differential equations 2024
Liyi Huang, Bowen Pang, Qiming Yang, **Xiangnan Feng**, Wei Wei
Knowledge-Based Systems 292, 111619
- A representation-learning-based approach to predict stock price trend via dynamic spatiotemporal feature embedding 2023
Bowen Pang, Wei Wei, Xing Li, **Xiangnan Feng**, Chao Li
Engineering Applications of Artificial Intelligence, Volume 126, Part A, 2023, 106849, ISSN 0952-1976
- Graphical representation and hierarchical decomposition mechanism for vertex-cover solution space 2023
Wei Wei, **Xiangnan Feng**
Applied Mathematics and Computation, Volume 458, 2023, 128264, ISSN 0096-3003
- The dynamic resilience of urban labour networks 2023
Xiangnan Feng, Alex Rutherford
Royal Society Open Science 10(7), 230214
- Representation learning of enhanced graphs using random walk graph convolutional network 2023
Xing Li, Wei Wei, Ruizhi Zhang, Zhenyu Shi, Zhiming Zheng, **Xiangnan Feng**
ACM Transactions on Intelligent Systems and Technology 2023 Feb 10
- Enhance ambiguous community structure via multi-strategy community related link prediction method with evolutionary process Under Review
Qiming Yang, Wei Wei, Ruizhi Zhang, Bowen Pang, **Xiangnan Feng**
arXiv:2204.13301
- Abstract: Shaping and Predicting the Urban Labor Markets 2021
Xiangnan Feng, Manuel Cebrian, Alex Rutherford
the 10th International Conference on Complex Networks and Their Applications, Madrid, Spain
- Representation learning of graphs using graph convolutional multilayer networks based on Motifs 2021
Xing Li, Wei Wei, **Xiangnan Feng**, Xue Liu, Zhiming Zheng
Neurocomputing, 2021, ISSN 0925-2312
- Effects of dynamic-Win-Stay-Lose-Learn model with voluntary participation in social dilemma 2021
Zhenyu Shi, Wei Wei, **Xiangnan Feng**, Ruizhi Zhang, Zhiming Zheng
Chaos, Solitons & Fractals, Volume 151, 2021, 111269, ISSN 0960-0779
- Graph classification based on skeleton and component features 2021
Xue Liu, Wei Wei, **Xiangnan Feng**, Xiaobo Cao, Dan Sun
Knowledge-Based Systems, Volume 228, 2021, 107301, ISSN 0950-7051
- Research of Motif-based similarity for link prediction problem 2021
Chao Li, Wei Wei, **Xiangnan Feng**, Jiaomin Liu
IEEE Access, vol. 9, pp. 66636-66645, 2021
- Dynamic aspiration based on Win-Stay-Lose-Learn rule in spatial prisoner's dilemma game 2021
Zhenyu Shi, Wei Wei, **Xiangnan Feng**, Xing Li, Zhiming Zheng
Plos one, 16(1), e0244814.
- Core influence mechanism on vertex-cover problem through leaf-removal-core breaking 2019
Xiangnan Feng, Wei Wei, Xing Li, Zhiming Zheng
Journal of Statistical Mechanics: Theory and Experiment, 2019.7 (2019): 073401
- Research on centralities based on von Neumann entropy for motifs 2019
Xiangnan Feng, Wei Wei, Zhiming Zheng
2019 International Conference on Artificial Intelligence and Computing Science
- Exploring the heterogeneity for node importance by von Neumann entropy 2018

Xiangnan Feng, Wei Wei, Renquan Zhang, Jiannan Wang, Ying Shi, Zhiming Zheng
Physica A: Statistical Mechanics and its Applications, Volume 517, 1 March 2019, Pages 53-65

Optimal stabilization of boolean networks through collective influence 2018
 Jiannan Wang, Sen Pei, Wei Wei, **Xiangnan Feng**, Zhiming Zheng
Physical Review E, 97, 032305 – Published 13 March 2018

Correlation research of centralities on complex network by statistical learning 2018
 Ying Shi, Wei Wei, **Xiangnan Feng**, Zhiming Zheng
2018 2nd International Conference on Artificial Intelligence and Software Engineering

Identifying influential vertices in boolean networks through dynamical voter rank 2017
 Jiannan Wang, **Xiangnan Feng**, Zhilong Mi, Ziqiao Yin, Zhiming Zheng
2017 IEEE 2nd Information Technology, Networking, Electronic and Automation Control Conference

Combined fit to BESIII data on $e^+e^- \rightarrow h_c\pi^+\pi^-$ and $\chi_{c0}\omega$ 2015
Xiangnan Feng, Xuyang Gao, Chengping Shen
International Journal of Modern Physics A, 30, 1550142 (2015)

Optimization model for malfunction detection in automatic lathe 2015
 Zhenfu Wang, Menglun Wang, Sen Chen, **Xiangnan Feng**
Modular Machine Tool & Automatic Manufacturing Technique, 2015, ISSN: 1001-2265 CN: 21-1132/TG

Photovoltaic hut design based on the greedy algorithm 2013
 Zhenfu Wang, Menglun Wang, Sen Chen, **Xiangnan Feng**
Acta Energiae Solaris Sinica, 2013 Vol. 34 (10): 1775-1780

Conferences and Presentations

Workshop Talk: Learning from a Software Task Space, Vienna 2025
 CSH Symposium, Vienna

Conference Talk: The Dynamic Resilience of Urban Labour Networks 2022
 Conference on Complex Systems 2022, Palma de Mallorca

Workshop Talk: Prediction the Future Labour Markets, Berlin 2021
 CHM Symposium, Berlin

Conference Talk: Shaping and Predicting the Urban Labor Markets 2021
 the 10th International Conference on Complex Networks and Their Applications, Madrid

Grants & Awards

Gemma 2 Academic Program GCP Credit Award (\$10,000), Google 2024 - 2025

EU Joint Research Center, Global hubs and local clusters in IT: a task-based analysis of the changing geography of software jobs 2025

Outstanding Graduate of BUAA 2021

Academic Excellence Foundation of BUAA, (\$3000, 85 out of 700) 2019

Activities

Reading Group: Mastering Metrics, Complexity Science Hub 2023
 Organize the reading group of book Mastering Metrics: The Path from Cause to Effect.

Volunteer in NetSci 2023, Vienna, Austria 2023

Seminar: Elements of Statistical Learning , Beihang University Organize the seminar of statistical learning as the group leader.	2017 - 2018
Overwatch Replay Analyzer (ORA) Develop the open-source software to extract a event timeline from computer game Overwatch video, which is used by professional Overwatch League E-Sports teams	2017 - 2018
Manager of Website: “Future Garden” , the Official BBS of Beihang University	2016 - 2020
BUAA University Tennis Team	2018, 2020
BUAA University Flag Football Team	2016
Internship in China Academy of Information and Communications Technology	07/2015 - 12/2015
Teaching Assistant in Calculus , Beihang University	09/2014 - 01/2015

Academic References

Frank Neffke Transforming Economies Group Leader Complexity Science Hub Vienna (CSH) ✉ neffke@csih.ac.at	Alex Rutherford Senior Data Scientist UK Communications Regulator Ofcom ✉ alexisadams@gmail.com	Manuel Cebrian Senior Research Scientist Spanish National Research Council ✉ manuel.cebrian@csic.es
Wei Wei Associate Professor School of Mathematical Sciences, Beihang University ✉ weiw@buaa.edu.cn		